

Lab 1 – Spoiler Alert Product Description

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1. Introduction

Food recalls represent a significant and ongoing public health concern in the United States. Allergen contamination, foreign materials, and foodborne illnesses like *Listeria* and *Salmonella* can impact millions of pounds of food products each year (Food Safety and Inspection Service, 2024). Although agencies such as the FDA and CDC do publish recall reports, a significant portion of consumers remain unaware of these reports. Only 13% of Americans take the time to access government websites for food recall information, and even fewer at 3% receive emails or texts about food recalls (Food Safety and Inspection Service, 2020). Food recalls are a continuing problem and without a reliable reporting system consumers will continue to be at risk.

Despite official government sources offering email notifications, it is outdated for the modern era of smartphone use. Other applications try to offer a means of reporting recalls, but they just are not customizable to individual concerns like specific allergies.

Despite the availability of recall notifications, there are challenges in how efficiently consumers receive, interpret, and respond to recall information. Nearly three quarters of consumers greatly underestimate the frequency of foodborne illness outbreaks, which indicates that a majority of consumers are not aware of unsafe food conditions (Food Safety and Inspection Service, 2020). Consumers tend to have an optimistic bias that food recalls are not relevant to them, believing that the products they purchase are less likely to be recalled (Hallman, Cuite, & Hooker, 2009). This bias can cause a lack of urgency in consumers with finding recall information and acting on recall reports. When learning about recalls, about four in ten consumers find recall information through media coverage while only about 6% get their information through sources like the FDA website (YouGov, 2025). A wide range of sources in which to find recall information leads to inconsistencies in how recall information is received.

Despite frequent recall announcements, about two fifths of consumers do not act on recall information and do not check their homes for recalled products (Hallman, Cuite, & Hooker, 2009).

These shortcomings are addressed by having a reliable, centralized location for recall information. Immediate and personalized alerts keep consumers motivated to stay informed. With a single platform that consolidates recall information into a single, accessible location, consumers will not have to navigate multiple sources to get important product recalls. Notifications will be personalized to the user, which increases relevance allows the consumer to focus on recalls that directly impact their household. With all relevant recalls located on one platform, consumers will not have to rely on media coverage or other sources to find recall reports. Consumers can be proactive about responding to recalls with alerts being delivered immediately. This proposed platform presents recalls in an urgent manner and promotes awareness and engagement.

Spoiler Alert is a recall notification application designed to improve the accessibility and relevance of food recall information. Spoiler Alert allows the user to keep a record of purchased products and create a profile that identifies allergy and dietary information to enable targeted recall notifications based on those individual risk factors. The user will be able to search for current and past recalls to motivate informed purchasing decisions. By continuously monitoring newly published recall reports, Spoiler Alert will be able to provide the user with recall notifications in real time. Spoiler Alert enhances efficiency and reliability of recall communication by addressing the current shortcomings of recall reporting.

2. Spoiler Alert Product Description

Spoiler Alert is a consumer-focused mobile application designed for consumer use that provides food recall information in one centralized location. The system integrates recall data from official government sources. Personalized notifications are sent to users based on individual profiles and records of purchased products. The application aims to improve the promptness of recall communication and enhance consumer awareness, accessibility, and response to recall information.

2.1 Key Product Features and Capabilities

Spoiler Alert collects recall and public health alert data from official government sources to provide a centralized recall database within the application. Recall records are retrieved from the openFDA API and the FSIS Recall API. The APIs return structured JSON data for recalls, such as product description, recall classification, and recall date. The retrieved data is then normalized for the Spoiler Alert database to support searching, filtering, and notification matching of recall data within the application.

Spoiler Alert users can create a personal account in which to store personal risk factors and product information within the system. Each user profile may include allergy information, dietary restrictions, and recently purchased products. This profile data is stored within the Spoiler Alert database which can then be used as filtering parameters when matching to recalls. When new recall data matches profile attributes, a personalized notification is generated for the affected user.

Spoiler Alert allows users to log purchased food items within the application. The items can be entered manually or by scanning the product barcode using the device's camera. When a barcode is scanned, the system queries a database of products using the item's UPC so product

details can be stored in the user profile. Products logged in the user profile can then be used to find relevant recall matches. The application also allows users to scan receipts to log multiple purchased items at once.

Spoiler Alert features an automated notification system which delivers relevant recall information to users in a timely manner. FDA and USDA recall data feeds are monitored for newly published recall reports. When the system retrieves a new report, data from the recall is matched against user profiles. If a match is found based on product information, allergens, or other criteria found within a user account, a personalized notification is generated. Generated notifications are sent directly to the user's device.

Spoiler alert has a search feature that allows users to find current and past recall reports. The user searches with keywords that the system uses to query the database of recalls and retrieves matching recall records. The search results contain detailed recall information that the user can use to make an informed decision before purchasing or consuming a product.

2.2 Major Components (Hardware/Software)

The Spoiler Alert application is compatible with both Android and iOS mobile devices. The application requires an active internet connection to retrieve recall data and deliver notifications. Mobile devices with a functioning camera will be able to utilize barcode and receipt scanning features for automated product logging.

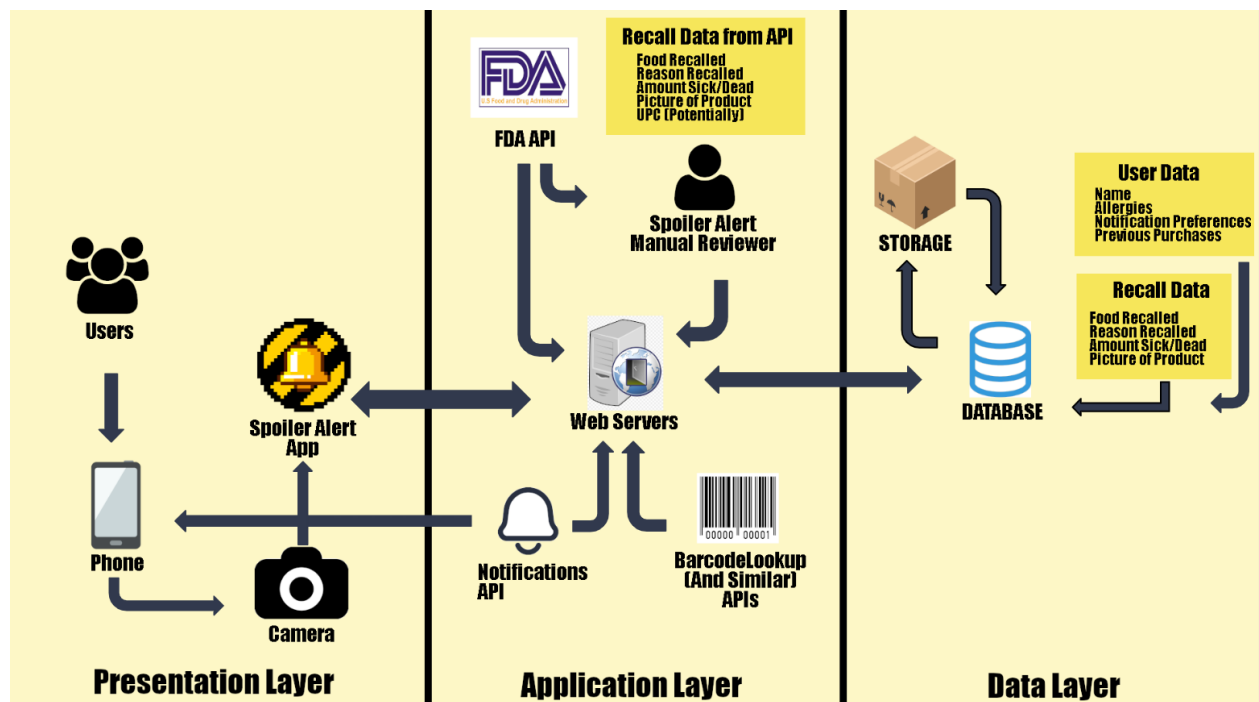
Consumers using the Spoiler Alert application will have the capability to create and build a personal profile. Profiles will include allergy information, dietary restrictions, notification preferences, and purchased products. These attributes are used when determining if a new recall is relevant to a specific user and how they receive the notification about the recall. Users will also have the capability to query the recall database directly.

Recalls and public health alerts are retrieved from the FDA openFDA API and the USDA FSIS Recall API. Retrieved data is returned in JSON format. The data is processed by the server to be normalized and stored in the database. Recall feeds are continuously monitored for new recall reports and public health alerts. The backend also supports barcode-based product lookup with the use of a barcode lookup API.

The database is structured to support search and filtering functionality and the recall matching process. User profiles and preferences, logged products, and recall records are stored within the relational database. Figure 1 illustrates the major functional components of the Spoiler Alert application.

Figure 1

Major Functional Components Diagram



3. Identification of Case Study

4. Glossary

Barcode Lookup API: The way Spoiler Alert is accessing FDA data.

CDC: Centers for Disease Control and Prevention

CPSC: U.S. Consumer Product Safety Commission

FDA: Food and Drug Administration

FSIS: Food Safety and Inspection Service, an agency of the United States Department of Agriculture

JSON: JavaScript Object Notation, a data interchange format used to store and transport data

openFDA API: The way Spoiler Alert is accessing FDA data.

Product Recall: When a manufacturer notifies the market of a product with a defect or potential danger.

UPC: Universal Product Code

USDA: U.S. Department of Agriculture

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[subsidiary-sentenced-connection-outbreak-salmonella-poisoning-related-peanut-butter](https://www.justice.gov/archives/opa/pr/conagra-subsidiary-sentenced-connection-outbreak-salmonella-poisoning-related-peanut-butter)

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